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Measures of Central Tendency
Or
Measures of Location

Definition of Measures of central tendency:

In a representative sample, the values of a series of data have a tendency to cluster around a certain point usually at the center of the series. This tendency of clustering the values around the center of the series is usually called tendency. And its numerical measures are called the measures of central tendency.

Or

A measure of central tendency is a typical value around which others figure congregate.

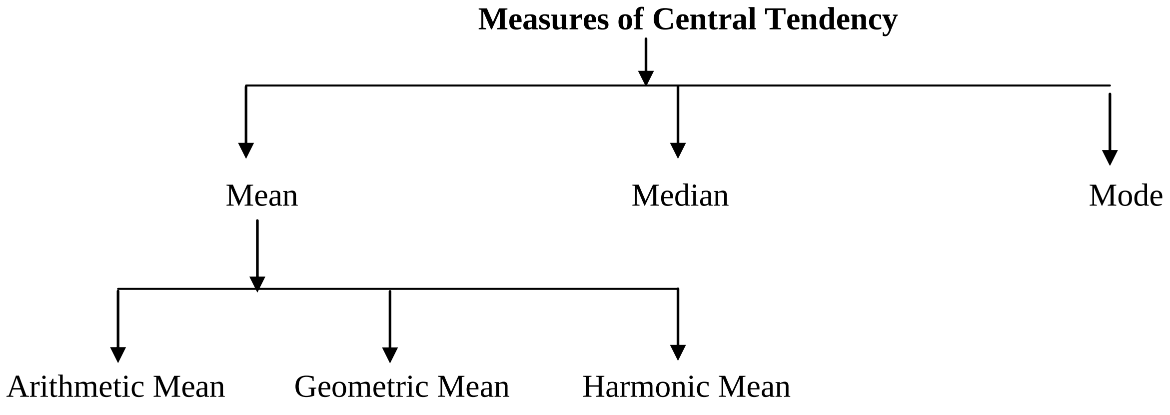
Objectives of Measures of central tendency:

1. To get one single value that describes the characteristics of the entire data.
2. To easily compare the data.

Types:

Different measures of central tendency are as follows:

1. Arithmetic Mean (A.M.)
2. Geometric Mean (G.M.)
3. Harmonic Mean (H.M.)
4. Median (Me)
5. Mode (Mo)



Arithmetic mean/ Average/Mean:

Arithmetic mean of a set of observations is the sum of all observations divided by the total number of observations. It is usually denoted by A.M or $\overline{X}$.
Or, the sum of a set of data divided by the number of data.

For ungroup data, let $x_{1,}x_{2,}x_{3,}\dots\ \dots\ \dots,x_n$ be the series of n observations in any statistical investigation.Then their arithmetic mean, $\overline{X} = \dfrac{x_1 + x_2 + \dots\dots + x_n}{n} = \dfrac{\sum_{i=1}^{n} x_i}{n}$

For group data, let $x_{1,}x_{2,}x_{3,}\dots\ \dots\ \dots,x_n$ be the series of n observations in any statistical investigation and $f_{1,}f_{2,}f_{3,}\dots\ \dots\ \dots,f_n$ be their corresponding frequencies.

Then their arithmetic mean, $\overline{X} = \dfrac{f_1x_1 + f_2x_2 + \dots\dots\dots + f_nx_n}{f_1 + f_2 + \dots\dots\dots + f_n} = \dfrac{\sum_{i=1}^{n} f_ix_i}{\sum_{i=1}^{n} f_i}$

Learn by heart:
Arithmetic mean:

For ungroup data	For group data
$\overline{X} = \dfrac{\sum_{i=1}^{n} x_i}{n}$	$\overline{X} = \dfrac{\sum_{i=1}^{n} f_ix_i}{\sum_{i=1}^{n} f_i}$

<u>Calculation of Arithmetic mean for ungroup data</u>

Example-1: Find the mean, from the following data:

9, 15, 11, 12, 3, 5, 10, 20, 14, 6, 8, 8, 12, 12, 18, 15, 6, 9, 18, 11

Mean / $\overline{X} = \dfrac{x_1 + x_2 + \dots\dots + x_n}{n}$

$$= \frac{9+15+11+12+3+5+10+20+14+6+8+8+12+12+18+15+6+9+18+11}{20} = \mathbf{11.1}$$

Example-2: Find the mean from the data: 5, 15, 10, 15, 5, 10, 10, 20, 25, 15.

Mean / $\overline{X} = \dfrac{x_1 + x_2 + \dots\dots x_n}{n}$

$$= \frac{5+15+10+15+5+10+10+20+25+15}{10} = \frac{130}{10} = \mathbf{13}$$

Practice-1: Find the mean from the data: 20, 37, 32, 39, 33, 34, 32

Practice-2: Find the mean from the data: 20.5, 37.5, 32.6, 39.9, 33.7, 34, 32.0

Practice-3: Find the mean from the data: 2, 7, -5, -7, 3.5, 4.7, 11

Calculation of arithmetic mean for group data

Example-1: Calculate arithmetic mean from the following grouped frequency table:

Number of games	Frequency fi
1 - 5	2
6 - 10	7
11 - 15	8
16 - 20	3

Solution: Estimating the mean from Grouped Data

Number of games	Frequency f	Midpoint x	$f_i x_i$
1 - 5	2	3	6
6 - 10	7	8	56
11 - 15	8	13	104
16 - 20	3	18	54
	$\sum f_i = 20$	Totals:	$\sum f_i X_i = 220$

We know, **Mean =** $\dfrac{\sum_{i=1}^{n} f_i x_i}{\sum_{i=1}^{n} f_i} = \dfrac{220}{20} = 11$

Example-2: Calculate arithmetic mean from the following data:

Length (mm)	Frequency
150 - 154	5
155 - 159	2
160 - 164	6
165 - 169	8
170 - 174	9
175 - 179	11
180 - 184	6
185 - 189	3

Solution: Calculating table

Length (mm)	Midpoint x	Frequency f	fx
150 - 154	152	5	760
155 - 159	157	2	314
160 - 164	162	6	972
165 - 169	167	8	1336
170 - 174	172	9	1548
175 - 179	177	11	1947
180 - 184	182	6	1092
185 - 189	187	3	561
	Totals:	$\sum f_i = \mathbf{50}$	$\sum f_i X_i = \mathbf{8530}$

We know, **Mean =** $\dfrac{\sum_{i=1}^{n} f_i x_i}{\sum_{i=1}^{n} f_i} = \dfrac{8530}{50} = 170.6$

Example-3-The ages of the 112 people who live on a tropical island were grouped as follows:

Age	Number
0 - 9	20
10 - 19	21
20 - 29	23
30 - 39	16
40 - 49	11
50 - 59	10
60 - 69	7
70 - 79	3
80 - 89	1

Solution: We know, **Mean =** $\dfrac{\sum_{i=1}^{n} f_i x_i}{\sum_{i=1}^{n} f_i}$

Calculate arithmetic mean from the above data:

Age	Midpoint x	Number f	fx
0 - 9	5	20	100
10 - 19	15	21	315
20 - 29	25	23	575

30 - 39	35	16	560
40 - 49	45	11	495
50 - 59	55	10	550
60 - 69	65	7	455
70 - 79	75	3	225
80 - 89	85	1	85
	Totals:	**112**	**3360**

Estimated Mean = $\frac{3360}{112}$ = **30**

Calculating Mean Using Short-cut Method

The short-cut method is suitable for grouped data. The formula is

$$\bar{x} = a + c\bar{d}$$

$$= a + \frac{\sum_{i=1}^{n} f_i d_i}{n} \times c$$ Where,

c = The size of class interval.
a = The assumed mean. (It is the middle no of the mid values).
$d_i = \frac{x_i - a}{c}$ The step deviation from a.
x_i = Mid values of each class.
n = The total frequency.

Example: Calculate mean for the following grouped data using short-cut method.

Class	0-10	10-20	20-30	30-40	40-50
Frequency	7	8	20	10	5

Solution: Table for calculating arithmetic mean by using short-cut method

Class	Mid-value (x_i)	Frequency (f_i)	$d_i = \frac{x_i - a}{c}$	$f_i d_i$
0-10	5	7	-2	-14
10-20	15	8	-1	-8
20-30	25 (a)	20	0	0
30-40	35	10	+1	+10
40-50	45	5	+2	+10
		n = 50		$\sum_{i=1}^{n} f_i d_i$ =-2

Here, a = 25 and c = 10

We know, mean $\bar{x} = a + c\bar{d}$ = a + $\frac{\sum_{i=1}^{n} f_i d_i}{n} \times c$ = 25 + $\frac{-2}{50} \times 10$ =24.6

For Practice: Calculate the arithmetic mean of the frequency distribution given below:

Height	130-134	135-139	140-144	145-149	150-154	155-159	160-164
No. of students	5	15	28	24	17	10	1

Properties of the Arithmetic Mean:

As stated, the mean is a widely used measure of location. It has several important properties.

1. Arithmetic mean depends on origin and scale of measurement.
2. The sum of the deviations of each value from the mean will always be zero, that is: $\boxed{\sum(X - \bar{X}) = 0}$
3. Every set of interval level and ratio level data has a mean.
4. The sum of the squares of the deviation of a set of observation is minimum when taken about mean. That is, $\sum_{i=1}^{n}(x_i - \bar{x})^2 \le \sum_{i=1}^{n}(x_i - a)^2$
5. All the data values are used in the calculation.
6. A set of data has only one mean, that is, the mean is unique.
7. The mean is a useful measure for comparing two or more populations.
8. Mean of composite series: If $\bar{X}_i, (i = 1, 2, \cdots\cdots, k)$ are the means of k-component series of sizes n_i, (i = 1, 2, ….., k) respectively, then the mean $\bar{X}$ of the composite series obtained on combining the component series is given by the formula:

$$\bar{X} = \frac{n_1\bar{X}_1 + n_2\bar{X}_2 + \cdots\cdots + n_k\bar{X}_k}{n_1 + n_2 + \cdots\cdots + n_k} = \frac{\sum_{i=1}^{k} n_i\bar{X}_i}{\sum_{i=1}^{k} n_i}$$

Merits: 1. It is rigidly defined.
2. It is easy to calculate and easy to understand.
3. It is based upon all the observations.
4. It is suitable for further algebraic treatment.
5. It is less affected by sampling fluctuations.

Demerits: 1. It is affected by extreme values.
2. It cannot be calculated if the extreme class is open.
3. It cannot be used when we are dealing with qualitative characteristics which cannot be measured quantitatively.

Geometric mean:

The geometric mean of a set of n non-zero positive observations is the nth root of their product.

For ungroup data, let $x_{1,}x_{2,}x_{3,}\dots\ \dots\ \dots,x_n$ be the series of n observations in any statistical investigation. Thus the geometric mean, G.M = $(x_1.x_2.\dots\dots\dots x_n)^{\frac{1}{n}}$

For group data, let $x_{1,}x_{2,}x_{3,}\dots\ \dots\ \dots,x_n$ be the series of n observations in any statistical investigation and $f_{1,}f_{2,}f_{3,}\dots\ \dots\ \dots,f_n$ be their corresponding frequencies.

Geometric mean, G.M = $(x_1^{f_1}.\ x_2^{f_2}\ \dots\dots\dots x_n^{f_n})^{\frac{1}{\sum f_i}}$

Learn by heart:

Geometric mean:

For ungroup data	For group data
G.M = $(x_1.x_2.\dots\dots\dots x_n)^{\frac{1}{n}}$	G.M = $(x_1^{f_1}.\ x_2^{f_2}\ \dots\dots\dots x_n^{f_n})^{\frac{1}{\sum f_i}}$

Geometric mean for ungroup data:

Example-1:

What is the geometric mean of 2, 8 and 4?

Solution: Multiply those numbers together. Then take the third root (cube root) because there are 3 numbers. G.M= $(2\times4\times8)^{\frac{1}{3}} = (64)^{\frac{1}{3}} = (4^3)^{\frac{1}{3}} = 4$

Example-2: To find the Geometric Mean of 1, 2, 3, 4, 5.

Step 1: N = 5, the total number of values. Find 1/N. 1/N = 0.2

Step 2: Geometric Mean = $((X_1)(X_2)(X_3)........(X_N))^{1/N}$

= $((1)(2)(3)(4)(5))^{0.2} = (120)^{0.2}$

So, Geometric Mean = 2.60517

Merits: 1. It is rigidly defined.
2. It takes all the observations into account.
3. It is used for further observations.
4. It is not affected by sampling fluctuations.

Demerits: 1. It is neither easy to calculate nor easy to be understand.
2. In case of odd number of negative class value, it cannot be computed at all.

Uses of GM:

i) To find the rate of population growth and the rate of interest
ii) In the construction of index number

Harmonic mean:

The harmonic mean of a set of n non zero observations $x_1.x_2........x_n$ in a series is the reciprocal of the arithmetic mean of the reciprocals.
For ungroup data, let $x_1, x_2, x_3, \dots \dots \dots, x_n$ be the series of n observations in any statistical investigation.

Thus, Harmonic Mean = $\dfrac{n}{\dfrac{1}{x_1}+\dfrac{1}{x_2}+\dots\dots\dots\dots+\dfrac{1}{x_n}} = \dfrac{n}{\sum_{i=1}^{n}\dfrac{1}{x_i}}$

For group data, let $x_1, x_2, x_3, \dots \dots \dots, x_n$ be the series of n observations in any statistical investigation and $f_1, f_2, f_3, \dots \dots \dots, f_n$ be their corresponding frequencies.

Harmonic mean, H.M = $\dfrac{n}{\sum_{i=1}^{n}\dfrac{f_i}{x_i}}$ Where n = $\sum_{i=1}^{n} f_i$

Learn by heart:
Harmonic mean:

For Ungrouped Data	For Grouped Data
$H.M \text{ of } X = \overline{X} = \dfrac{n}{\Sigma\left(\dfrac{1}{x}\right)}$	$H.M \text{ of } X = \overline{X} = \dfrac{\Sigma f}{\Sigma\left(\dfrac{f}{x}\right)}$

Solved Examples-1:

Find the harmonic mean of the following data 8, 9, 6, 11, 10, 5
Solution:

Given data: 8, 9, 6, 11, 10, 5.

So Harmonic mean = $\dfrac{6}{\dfrac{1}{8}+\dfrac{1}{9}+\dfrac{1}{6}+\dfrac{1}{11}+\dfrac{1}{10}+\dfrac{1}{5}} = \dfrac{6}{0.7396} = 7.560$

Exercise-1: Find the harmonic mean of the following data 90, 35, 45, 76, 58, 37, 87.

Example-2:
Calculate the harmonic mean of the numbers: 13.2, 14.2, 14.8, 15.2 and 16.1Solution:
The harmonic mean is calculated as below:

x	$\dfrac{1}{x}$
13.2	0.0758
14.2	0.0704
14.8	0.0676
15.2	0.0658

16.1	0.0621
Total	$\Sigma\left(\frac{1}{x}\right)=0.3417$

$$H.M \text{ of } X=\overline{X}=\frac{n}{\Sigma\left(\frac{1}{x}\right)}$$

$$H.M \text{ of } X=\overline{X}=\frac{5}{0.3417}=14.63$$

Example-1: Calculate the harmonic mean for the given below:

Marks	$30-39$	$40-49$	$50-59$	$60-69$	$70-79$	$80-89$	$90-99$
f	2	3	11	20	32	25	7

Solution:
The necessary calculations are given below:

Marks	x	f	$\frac{f}{x}$
$30-39$	34.5	2	0.0580
$40-49$	44.5	3	0.0674
$50-59$	54.5	11	0.2018
$60-69$	64.5	20	0.3101
$70-79$	74.5	32	0.4295
$80-89$	84.5	25	0.2959
$90-99$	94.5	7	0.0741
Total		$\sum f=100$	$\Sigma\left(\frac{f}{x}\right)=1.4368$

Now we will find the Harmonic Mean as

$$\overline{X}=\frac{\sum f}{\Sigma\left(\frac{f}{x}\right)}=\frac{100}{1.4368}=69.60$$

Merits: 1. It is rigidly defined.
2. It is based upon all the observations.

3. It is used for further algebraic treatment.
4. It is not affected by sampling fluctuations.

Demerits: 1. It is not easy to understand and is difficult to calculate.
2. It can be calculated if the extreme class is open.

Uses of HM:
It is used for calculating average speed of automobiles.

Median:

The median is defined as the middle most observation when the observations are arranged in order of magnitude.

Calculation of Median (Ungrouped Data)

- First arrange them in ascending or descending order and count number of observation or items n.
- When n is odd, Median = $\left(\frac{5+1}{2}\right)$ th observation.
- When n is even, Median = $\dfrac{\frac{n}{2}+\left(\frac{n}{2}+1\right)}{2}$ th observation.

Calculation of Median (grouped Data)

Median = L + $\dfrac{\frac{n}{2}-F}{f}$ x c

Where, L =Lower limit of the median class
n = Total number of observations.
F = Cumulative frequency of the class just proceeding the median class.
f = Frequency of the median class.
c = length of the median class

Learn by heart:
Median:

For ungroup data	For group data
When n is odd, Median = $\left(\frac{n+1}{2}\right)$ th observation. When n is even, Median = $\dfrac{\frac{n}{2}+\left(\frac{n}{2}+1\right)}{2}$ th	Median = L + $\dfrac{\frac{n}{2}-F}{f}$ x c

<u>**Calculation of median For ungrouped data**</u>

<u>When number of observations (n) is odd</u>

Example-1: Find the median for the following data: 15, 10, 13, 5, 20, 25, and 17.

Solution: Firstly we arranged the observations by ascending order of magnitude:

5, 10, 13, **15,** 17, 20, 25

From the above observation we have seen that total number observation is 7, which is odd.

So median = $\left(\frac{n+1}{2}\right)$ th observation.

$= \left(\frac{7+1}{2}\right)$

= 4' th observation.

= 15

<u>When number of observations (n) is even</u>

Example-2: Find the median for the following data: 15, 10, 13, 5, 20, 25

Solution: Firstly we arranged the observations by ascending order of magnitude:

5, 10, 13, 15, 20, 25

From the above observation we have seen that total number observation is 6, which is even.

So median = $\dfrac{\frac{n}{2}+\left(\frac{n}{2}+1\right)}{2}$ th observation.

$= \dfrac{\frac{6}{2}th+\left(\frac{6}{2}+1\right)th}{2}$ observation

$= \dfrac{3thobservation+4thobservation}{2}$

$= \dfrac{13+15}{2}$ = 14

<u>For practice-1:</u> Find the median for the following data: 25, 1, 3, 5, 2, 5, 9, and 22

<u>For practice-2:</u> Find the median for the following data: 5, 1, 3, 11, 2, 0, and 7

<u>Estimating the Median from Grouped Data</u>

Example-1: Calculate median from the following data

Number of games	**Frequency**

150 - 155	5
155 - 160	2
160 - 165	6
165 - 170	8
170 - 175	9
175 - 180	11
180 - 185	6
185 - 190	3

Solution: Calculation of median

Number of games	**Frequency (f)**	**Cumulative frequency (F.C)**
150 - 155	5	5
155 - 160	2	7
160 - 165	6	13
165 - 170	8	21
170 - 175	**9**	**30**
175 - 180	11	41
180 - 185	6	47
185 - 190	3	50
	$\sum f_i = \mathbf{50}$	

For group data, Median = L + $\dfrac{\frac{n}{2} - F_c}{f_m}$ x c = 170 + $\dfrac{25-21}{9}$ x 5 = 172.22

Here,
(i) n =50 And n/2 = 50/2 = 25
(ii) Since n = 25 we have seen from the table median class is 170-175
(iii) L = 170
(iv) f_m = 9
(v) C = 5 and
(vi) F_c = 21

Exercise-1: The ages of the 112 people who live on a tropical island were grouped as follows:

Age-inclusive	**Number**
0 - 9	20
10 - 19	21
20 - 29	23
30 - 39	16
40 - 49	11

50 - 59	10
60 - 69	7
70 - 79	3
80 - 89	1

Calculate median from the above data

Merits: 1. It is rigidly defined.
2. It is easy to understand and easy to calculate.
3. It is not affected by extreme observations.
4. It can be calculated form frequency distribution, with end class.

Demerits: 1. It is not based on all observations.
2. It is not suitable for further algebraic treatment.
3. It is affected by sampling fluctuations.

Uses of Median:

i) It is the only average to be used while dealing with qualitative data, which cannot be measured quantitatively but still can be arranged in ascending or descending order of magnitude. e.g., to find the average intelligence, or average honesty among a group of people.
ii) It is to be used to determining the typical values in the problems concerning wages, distribution of wealth, etc.

Properties of the Median:

The major properties of the median are:

1. The median is a unique value, that is, like the mean, there is only one median for a set of data.
2. It is not influenced by extremely large or small values.
3. It can be computed for ratio level, interval level, and ordinal-level data.
4. Fifty percent of the observations are greater than the median and fifty percent of the observations are less than the median.

Mode:

The mode is that observation of the variable for which the frequency is maximum.
For example, the mode of the observations 2, 5, 9, 5, 3, 5 is 5

For group data, Mode= $L + \frac{\Delta_1}{\Delta_1 + \Delta_2} \times c$

Where, L = Lower limit of the modal class.
Δ_1 = Difference between frequency of the modal class and pre-modal class.
Δ_2 = Difference between frequency of the modal class and post-modal class.
c = length of the modal class.

Learn by heart:

Mode:

For ungroup data	**For group data**

Maximum frequency	Mode= L + $\frac{\Delta_1}{\Delta_1 + \Delta_2}$ x c

Example-2: Find mode for the following data: 5, 15, 10, 15, 5, 10, 10, 20, 25, 15.

Mode: Two numbers appear **most often: 10 and 15.** There are three 10's and three 15's. In this example there are two answers for the mode. So mode is not unique.

Example-2: Find mode from the following data:

9, 15, 11, 12, 3, 5, 10, 20, 14, 6, 8, 8, 12, 12, 18, 15, 6, 9, 18, 11

The Mode is the number which appears most often (you can have more than one mode):

3, 5, 6, 6, 8, 8, 9, 9, 10, 11, 11, 12, 12, 12, 14, 15, 15, 18, 18, 20:

12 appears three times, more often than the other values, so **Mode = 12**

Estimating the Mode from Grouped Data

Example-2: Calculate mode from the following data:

Length (mm)	**Frequency**
150 - 154	5
155 - 159	2
160 - 164	7
165 - 169	10
170 - 174	9
175 - 179	8
180 - 184	6
185 - 189	3

Solution: Calculation of mode:

Number of games	**Frequency (f)**
150 - 155	5
155 - 160	2
160 - 165	7
165 - 170	**10**
170 - 175	9
175 - 180	8
180 - 185	6
185 - 190	3

For group data, Mode= L + $\frac{\Delta_1}{\Delta_1 + \Delta_2}$ x c

$$= 165 + \frac{3}{3+1} \times 5 = 168.75\text{mm}$$

Here, modal class is 165-17 (Maximum frequency), L = 165 and C = 5

$\Delta_1 = 10 - 7$

$\Delta_2 = 10 - 9$

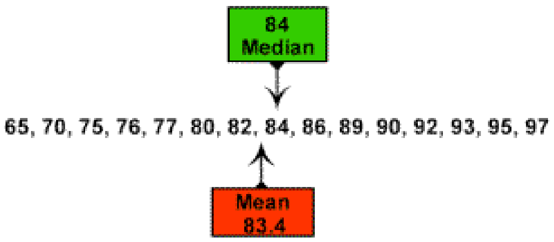
Exercise-2: The ages of the 112 people who live on a tropical island were grouped as follows:

Age-Inclusive	**Number**
0 - 9	20
10 - 19	21
20 - 29	23
30 - 39	16
40 - 49	11
50 - 59	10
60 - 69	7
70 - 79	3
80 - 89	1

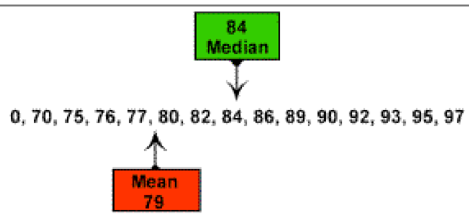
Calculate mode from the above data

Example:

Consider this set of test score values:



Normal listing of scores.



Scores with the lowest score replaced with outlier.

Example:

What will happen to the measures of central tendency if we add the same amount to all data values, or multiply each data value by the same amount?

	Data	Mean	Mode	Median
Original Data Set:	6, 7, 8, 10, 12, 14, 14, 15, 16, 20	12.2	14	13
Add 3 to each	9. 10. 11. 13. 15. 17. 17. 18. 19. 23	15.2	17	16

data value				
Multiply 2 times each data value	12, 14, 16, 20, 24, 28, 28, 30, 32, 40	24.4	28	26

Exercises 1–6, find the mean,GM, HM, median, and mode of the set of measurements.

1. 5, 12, 7, 14, 8, 9, 7
2. 30, 37, 32, 39, 33, 34, 32
3. 5, 12, 7, 24, 8, 9, 7
4. 20, 37, 32, 39, 33, 34, 32
5. 5, 12, 7, 14, 9, 7
6. 30, 37, 32, 39, 34, 32
7.-2, 3, 0, 1-, 7, 0, 7,3,7

Merits : 1. It is easy to understand and easy to calculate.
2. It is not affected by extreme value.
3. It can be calculated from frequency distribution with open class.

Demerits: 1. It is affected by sampling fluctuation.
2. It is not based upon all the observations.
3. It is not suitable for further algebraic treatment.
4. It is not clearly defined in case of multimodal distribution.

Properties of the Mode:

i) The mode can be found for all levels of data (nominal, ordinal, interval, and ratio).
ii) The mode is not affected by extremely high or low values.
iii) A set of data can have more than one mode. If it has two modes, it is said to be bimodal.
iv) A disadvantage is that a set of data may not have a mode because no value appears more than once.

Empirical Relation between Mean, Median, Mode

$$\boxed{\textbf{Mode = 3 Median – 2 Mean}}$$

Other Location Measures:

Median and mode are widely used two location measures. In addition to median and mode there are other location measures also. They are

i) Quartiles, ii) Deciles, iii) Percentiles

Quartiles: Quartiles are those values which divide the total frequency into four parts. We need three values to divide the whole frequency into four parts. That is why there are three quartile Q_1 denote first quartile, Q_2 second quartile, Q_3 third quartile. **is called the median of the frequency.**

Q_1 Q_2 Q_3

The quartiles are important in grading, rating, scoring, ranking etc.

Calculation of Quartiles (Grouped Data)

For Grouped data, quartiles $Q_i = \quad L + \dfrac{\dfrac{i \times n}{4} - F_c}{f_Q} \times c \qquad ; i = 1,2,3$

Where
C = The size of class interval.
L = Lower limit of quartile class.
F_c= Preceding cumulative frequency of quartile class.
f_Q= Frequency of the quartile class.

Deciles: Deciles are those values which divide the total frequency into ten parts. We need nine values to divide the whole frequency into ten parts. Deciles are denoted by D_1, D_2…….D_9. D_1 indicate first deciles, D_2 indicate second deciles etc.

Calculation of Deciles (Grouped Data)

For Grouped data, deciles $D_i = \quad L + \dfrac{\dfrac{i \times n}{10} - F_c}{f_D} \times c \qquad ; i = 1,2,3........,9$

Percentiles: Percentiles are those values which divide the total frequency into hundred parts. We need ninety nine values to divide the whole frequency into hundred parts. Percentiles are denoted by $\mathbf{P_1}$, $\mathbf{P_2}$**…..** $\mathbf{P_{15}}$**……………..**$\mathbf{P_{99}}$, etc.

Calculation of Percentiles (Grouped Data)

For Grouped data, percentiles P_i = L + $L + \dfrac{\dfrac{i \times n}{100} - F_c}{f_p} \times c \qquad ; i = 1,2,3........,99$

Weighted Mean

The ***weighted mean*** is a special case of the arithmetic mean. It is often useful when there are several observations of the same value.

> ***Weighted mean***: The value of each observation is multiplied by the number of times it occurs. The sum of these products is divided by the total number of observations to give the weighted mean.

In general, the weighted mean of a set of numbers, designated $X_1, X_2, X_3, \ldots X_n$, with the corresponding weights $w_1, w_2, w_3, \ldots, w_n$ is computed by:

Weighted Mean $$\overline{X}_w = \frac{w_1X_1 + w_2X_2 + w_3X_3 + \cdots + w_nX_n}{w_1 + w_2 + w_3 + \cdots + w_n} \qquad [3-3]$$

The weighted mean is particularly useful when various classes or groups contribute differently to the total. For example, the coronary care unit of a hospital consists of nurses= aides who are paid \$12 per hour, nurses = assistants who earn \$15 per hour, and registered nurses who earn \$21 per hour.
It would not be accurate to say the average hourly wage for the coronary unit is \$16 per hour (\$12 + \$15 + \$21) / 3 unless there was the same number of people in each group.

Suppose the coronary care unit has ten employees: two aides who earn \$12 per hour, 3 nurses= assistants who earn \$15 per hour, and five registered nurses who earn \$21 per hour. The weighted mean is:

$$\overline{X}_w = \frac{w_1X_1 + w_2X_2 + w_3X_3 + \cdots + w_nX_n}{w_1 + w_2 + w_3 + \cdots + w_n}$$

$$= \frac{(2\times\$12)+(3\times\$15)+(5\times\$21)}{2+3+5} = \frac{\$24+\$45+\$105}{10} = \frac{\$174}{10} = \$17.40$$

Thus the weighted mean is \$17.40.

Characteristics / Criteria of ideal measures of Central Tendency. Which is the best measure of central tendency and why?

Ans: Characteristics of ideal measures of central tendency are given below:

1. It should be rigidly defined.
2. It should easy to calculate and easy to understand.
3. It should be based upon all observations.
4. It should be suitable for further algebraic treatments.
5. It should less affected by sampling fluctuation.
6. It should be least affected by extreme observation.

Since arithmetic mean satisfies most of the criteria given above, it's considered as the best of central tendency.

Theorem: For two non-zero positive observations show that,

(i) A.M ≥ G.M ≥ H.M

(ii) When A.M = G.M = H.M

(iii) A.M × H.M = GM^2

or

$\sqrt{\mathbf{A.M \times G.M}}$ = G.M

Ans. (1) Let x_1 and x_2 be two non-zero positive observations. then their arithmetic mean:

A.M = $\frac{x_{1+}x_2}{2}$

G.M = $(x_1.x_2)^{\frac{1}{2}} = \sqrt{x_1.x_2}$

H.M = $\frac{2}{\frac{1}{x_1}+\frac{1}{x_2}}$

Since, x_1 and x_2 are two non-zero positive observations then

$$(\sqrt{x_1} - \sqrt{x_2})^2 \geq 0$$

Or, $(\sqrt{x_1})^2 - 2.\sqrt{x_1}.\sqrt{x_2} + (\sqrt{x_2})^2 \geq 0$

Or, $x_1 - 2\sqrt{x_1x_2} + x_2 \geq 0$

Or, $x_1 + x_2 \geq 2\sqrt{x_1x_2}$

$$\textbf{Or, } \frac{x_1+x_2}{2} \geq \sqrt{x_1 x_2}$$

Or, A.M ≥ G.M … … …(1)

And Now,

$$\left(\frac{1}{\sqrt{x_1}} - \frac{1}{\sqrt{x_2}}\right)^2 \geq 0$$

$$\textbf{Or, } \left(\frac{1}{\sqrt{x_1}}\right)^2 - 2.\frac{1}{\sqrt{x_1}}.\frac{1}{\sqrt{x_2}} + \left(\frac{1}{\sqrt{x_2}}\right)^2 \geq 0$$

$$\textbf{Or, } \frac{1}{x_1} - \frac{2}{\sqrt{x_1 x_2}} + \frac{1}{x_2} \geq 0$$

$$\textbf{Or, } \frac{1}{x_1} + \frac{1}{x_2} \geq \frac{2}{\sqrt{x_1 x_2}}$$

$$\textbf{Or, } \sqrt{x_1 . x_2} \geq \frac{2}{\frac{1}{x_1}+\frac{1}{x_2}}$$

Or, G.M ≥ H.M … … …(2)

From equation (1) and (2) we can write, A.M ≥ G.M ≥ H.M. $[Proved]$

(2) If $x_1 = x_2 = x$

Or, $x_1 = x_2 = a$ then A.M = G.M = H.M.

Verification:

$$\textbf{A.M} = \frac{x_1+x_2}{2} = \frac{x+x}{2} = \frac{2x}{2} = x$$

$$\textbf{G.M} = (x_1 . x_2)^{\frac{1}{2}} = (x.x)^{\frac{1}{2}} = (x^2)^{\frac{1}{2}} = x .$$

$$\textbf{H.M} = \frac{2}{\frac{1}{x_1}+\frac{1}{x_2}} = \frac{2}{\frac{1}{x}+\frac{1}{x}} = \frac{2}{\frac{2}{x}} = x .$$

So, A.M = G.M = H.M. $[\boldsymbol{Proved}]$

(3) Let x_1 and x_2 be two non-zero positive observations then,

$$\textbf{A.M} \times \textbf{H.M} = \frac{x_1+x_2}{2} \times \frac{2}{\frac{1}{x_1}+\frac{1}{x_2}}$$

$$= \frac{x_1+x_2}{2} \times \frac{2}{\frac{x_1+x_2}{x_1x_2}}$$

$$= \frac{x_1+x_2}{2} \times 2 \times \frac{x_1x_2}{x_1+x_2}$$

$$= x_1x_2$$

$$= (\sqrt{x_1x_2})^2$$

$$= \text{GM}^2 \text{ . } [Proved]$$

Assignmen-1:

The numbers of incorrect answers on a true-false competency test for a random sample of 15 students were recorded as follows: 2, 1, 3, 0, 1, 3, 6, 0, 3, 3, 5, 2, 1, 4, and 2.

Find (i) Mean, median and Mode from the above data.

Assignmen-2:

The profits (in thousands) in a particular month earned by a number of salesmen of a large company are given below:

Class intervals	Frequency
0-10	10
10-20	20
20-30	35
30-40	40
40-50	25
50-60	25
60-70	15

(i) Calculate average profit
(ii) Calculate mode and median
(iii) Locate mode graphically
(iv) Locate median graphically

H.W: Which measures of central tendency is the best and why?